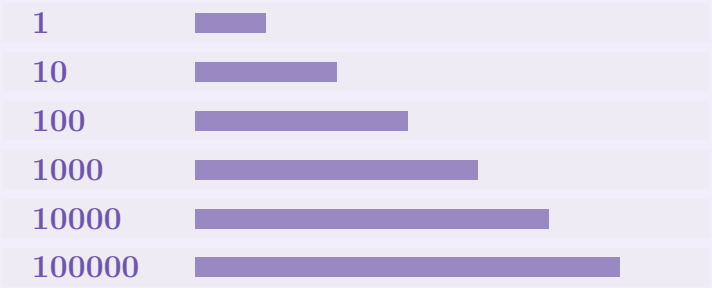
GANITA PRAKASH

CLASS 7 MATHEMATICS | PART I

LARGE NUMBERS AROUND US

Solved textbook exercises + illustrated notes



Questions first → worked explanations → visual notes → revision

Aligned with the supplied NCERT *Ganita Prakash*, Grade 7, Part I.
Original explanations and original TikZ illustrations; editable in Overleaf.

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Textbook exercises: worked answers first

How to study this section

The first pages answer common and important textbook prompts directly. **After these worked answers, the full additional solution text supplied with the chapter is included, organised by original textbook page.** For figure-based questions, keep the textbook's diagram beside these notes.

Textbook question + answer

Textbook p. 2–3. Fill the number ladder from 999 to a lakh and answer the Chintamani population questions. **Answer:** $999 + 1 = 1000$, largest four-digit number 9999, smallest five-digit number 10,000, largest five-digit 99,999, next 1,00,000. Population shortfall: $1{,}00{,}000 - 75{,}000 = 25{,}000$. Excess in 2024: $1{,}06{,}000 - 1{,}00{,}000 = 6{,}000$. Increase: $1{,}06{,}000 - 75{,}000 = 31{,}000$.

Textbook question + answer

Textbook p. 2. Could a person taste 1,00,000 rice varieties in 100 years? **Answer:** Ignoring leap years, $100 \times 365 = 36{,}500$ days. At one variety daily: 36,500; two: 73,000 (not enough); three: 1,09,500 (enough). **Assumption:** one different variety per tasting.

Textbook question + answer

Textbook p. 4–5. Convert between Indian place-value names and numerals. **Answer:** **In words:** 3,00,600: three lakh six hundred; 5,04,085: five lakh four thousand eighty-five; 27,30,000: twenty-seven lakh thirty thousand; 70,53,138: seventy lakh fifty-three thousand one hundred thirty-eight. **In numerals:** one lakh twenty-three thousand four hundred fifty-six = 1,23,456; four lakh seven thousand seven hundred four = 4,07,704; fifty lakh five thousand fifty = 50,05,050; ten lakh two hundred thirty-five = 10,00,235.

Textbook question + answer

Textbook p. 5–6. How many presses on a calculator having only +1000, +100, or +10? **Answer:** Target divided by button value. 53,000 needs 53 presses of +1000, 530 presses of +100, or 5300 presses of +10. One lakh requires 100, 1000, or 10,000 presses respectively. A +100 calculator **cannot** make a number that a +10 calculator cannot make.

Textbook question + answer

Textbook p. 6–8. Creative Chitti and Systematic Sippy: represent 5072, 8300 with button clicks. **Answer:** $5072 = 5 \times 1000 + 7 \times 10 + 2 \times 1$ (**14 clicks**); $8300 = 8 \times 1000 + 3 \times 100$ (**11 clicks**). To minimise clicks using powers of ten, press the largest available place-value buttons as often as needed.

Textbook question + answer

Textbook p. 6–7. Give two different valid representations of 3,67,813. **Answer:** $3 \times 100000 + 6 \times 10000 + 7 \times 1000 + 8 \times 100 + 1 \times 10 + 3$; also $36 \times 10000 + 78 \times 100 + 13$. **Check:** $360{,}000 + 7800 + 13 = 367{,}813$. Some supplied solution text includes an erroneous alternative $36 \times 10000 + 78 \times 1000 + 13$ (which equals 438,013). Do not copy that error.

Textbook question + answer

Textbook p. 7. Exactly 30 button presses for a three-digit number: find extremes. **Answer:** A possible maximum is 993: $9 \times 100 + 8 \times 10 + 13 \times 1 = 993$ uses $9 + 8 + 13 = 30$ presses. A possible minimum is 102: $10 \times 10 + 2 \times 1 = 102$ uses 12, not 30 presses; replace two of those ten-clicks by

twenty-one-click equivalents (*example:* $8 \times 10 + 22 \times 1 = 102$, $8 + 22 = 30$). Hence the extremes are 993 and 102.

Textbook question + answer

Textbook p. 19–20. Reality checks and estimation. **Answer:** Travelling 100 km/day for 10 years gives $100 \times 365 \times 10 = 365{,}000$ km, less than the quoted Moon distance 384,400 km. One million coins at one per second takes $1{,}000{,}000/86{,}400 \approx 11.57$ days, not a day. A billion equals 100 crore $= 10{,}000$ lakh.

Textbook question + answer

Textbook p. 21. Statue of Unity coin stack and migratory birds. **Answer:** If a coin is 1 mm thick, a 180 m stack needs 180,000 coins. A 12,000 km flight at 900–1000 km/day takes roughly 12–14 days. Godwit: $13{,}560/11 \approx 1233$ km/day, about 51.4 km/hour. A 40 m building fits $180/40 = 4.5$ times into the statue's height.

Common Mistake

Be careful: 100 lakh = 1 crore, but 10,000 **lakh** = 1 **billion**. Also, a word problem that asks for an estimate should not be answered with unwarranted exactness.

Further textbook exercise solutions (editable text)

How to use this companion answer section

The supplied chapter PDF contains *additional solution pages after the textbook chapter*. The following is an **editable transcription** organised in that same order, so that page-based questions are not lost. Equations originally set as stacked fractions, tables, and picture puzzles may lose some spatial formatting in automatic text extraction. **For those, consult the original supplied PDF figure**; the worked explanations above and the visual notes later are the clearest first reference. Answers in the supplied companion have not all been independently verified.

More textbook solutions — companion p. 1

Page No. 3

Figure it out

1. According to the 2011 Census, the population of the town of Chintamani was about 75,000. How much less than one lakh is 75,000?

Answer: 75,000 is 25,000 less than 1 lakh.

2. The estimated population of Chintamani in the year 2024 is 1,06,000. How much more than one lakh is 1,06,000?

Answer: 1,06,000 is 6,000 more than 1 lakh.

3. By how much did the population of Chintamani increase from 2011 to 2024?

Answer: Increase in population from 2011 to 2024 is 31,000. Look at the picture on the right. Somu is 1 metre tall. If each floor is about four times his height, what is the approximate height of the building? (Refer textbook for the picture)

Answer: 40 m Which is taller — The Statue of Unity or this building? How much taller? _____m.

Answer: The Statue of Unity is 140 m taller than the building. How tall is the Kunchikal waterfall compared to the building? 410 m. How many floors should Somu's building have to be as high as the waterfall? approx 113 floors .

Page No. 4 Write each of the numbers given below in words: (a) 3,00,600

Answer: Three lakh six hundred (b) 5,04,085

Answer: Five lakh four thousand eighty five

More textbook solutions — companion p. 2

Answer: Twenty seven lakh thirty thousand (d) 70,53,138

Answer: Seventy lakh fifty three thousand one hundred thirty eight

Page No. 5 Write the corresponding number for each of the following: (a) One lakh twenty-three thousand four hundred and fifty-six.

Answer: 1,23,456 (b) Four lakh seven thousand seven hundred and four.

Answer: 4,07,704 (c) Fifty lakhs five thousand and fifty.

Answer: 50,05,050 (d) Ten lakhs two hundred and thirty-five.

Answer: 10,00,235 Thoughtful thousand (a) Three thousand? 3 times (b) 10,000? 10 times (c) Fifty three thousand? 53 times (d) 90,000? 90 times (e) One Lakh? 100 times (f) 1,53,000 ? 153 times (g) How many thousands are required to make one lakh? 100 Tedious tens (a) Five hundred? 50 times (b) 780? 78 times (c) 1000? 100 times (d) 3700? 370 times (e) 10,000? 1000 times (f) One lakh? 10,000 times

More textbook solutions — companion p. 3

Answer: No

Page No. 6 Find a different way to get 5072 and write an expression for the same.

Buttons 5072 +10,00,000 +1,00,000 +10,000 +1,000 5 4 1 +100 10 40 +10 7 7 6 +1 2 2 12

More textbook solutions — companion p. 4

Figure it out

1. For each number given below, write expressions for at least two different ways to obtain the number through button clicks. Think like Chitti and be creative.

Answer: Some ways are given below. Try more ways. (a) 8300

Answer: $(8 \times 1000) + (3 \times 100)$ or, $(5 \times 1000) + (33 \times 100)$ (b) 40629

Answer: $(40 \times 1000) + (6 \times 100) + (2 \times 10) + (9 \times 1)$ or, $(30 \times 1000) + (106 \times 100) + (29 \times 1)$ (c) 56354

Answer: $(56 \times 1000) + (3 \times 100) + (5 \times 10) + (4 \times 1)$ or, $(46 \times 1000) + (103 \times 100) + (54 \times 1)$ (d) 66666

Answer: $(66 \times 1000) + (6 \times 100) + (66 \times 1)$ or, $(6 \times 10000) + (6 \times 1000) + (6 \times 100) + (6 \times 10) + (6 \times 1)$ (e) 367813

Answer: $(3 \times 100000) + (6 \times 10000) + (7 \times 1000)$) $+ (8 \times 100) + (1 \times 10) + (3 \times 1)$ or, $(36 \times 10000) + (78 \times 100) + (13 \times 1)$

Page No. 7 Creative Chitti has some questions for you — (a) You have to make exactly 30 button presses. What is the largest 3- digit number you can make? What is the smallest 3-digit number you can make?

Answer: 993 is the largest 3-digit number that can be made using 30 button presses. 102 is the smallest 3-digit number that can be made using 30 button presses.

More textbook solutions — companion p. 5

Answer: 997 can also be made as – $(8 \times 100) + (19 \times 10) + (7 \times 1)$ Number of clicks $= 8 + 19 + 7 = 34$ How can we get the numbers (a) 5072, (b) 8300, using as few button clicks as possible?

Answer: (a) $5072 = (5 \times 1000) + (7 \times 10) + (2 \times 1)$ Number of clicks $= 14$ (b) $8300 = (8 \times 1000) + (3 \times 100)$ Number of clicks $= 11$

Figure it out

1. For the numbers in the previous exercise, find out how to get each number by making the smallest number of button clicks and write the expression.

Answer: (a) $8300 = (8 \times 1000) + (3 \times 100)$ Number of clicks $= 11$ (b) $40629 = (4 \times 10000) + (6 \times 100) + (2 \times 10) + (9 \times 1)$ Number of clicks $= 4 + 6 + 2 + 9 = 21$ (c) $56354 = (5 \times 10000) + (6 \times 1000) + (3 \times 100) + (5 \times 10) + (4 \times 1)$ Number of clicks $= 5 + 6 + 3 + 5 + 4 = 23$ (d) $66666 = (6 \times 10000) + (6 \times 1000) + (6 \times 100) + (6 \times 10) + (6 \times 1)$ Number of clicks $= 6 + 6 + 6 + 6 + 6 = 30$ (f) $367813 = (3 \times 100000) + (6 \times 10000) + (7 \times 1000) + (8 \times 100) + (1 \times 10) + (3 \times 1)$ Number of clicks $= 3 + 6 + 7 + 8 + 1 + 3 = 28$

More textbook solutions — companion p. 6

2. Do you see any connection between each number and the corresponding smallest number of button clicks?

Answer: The smallest number of buttons click for each number is the sum of its digit.

Page No. 8 How many zeros does a thousand lakh have? 8 zeros How many zeros does a hundred thousand have? 5 zeros

Page No. 9

Figure it out

1. Read the numbers and write their number names in both the Indian and American systems: Sl. No. Indian System American System (a) 4050678 40,50,678 Forty Lakh Fifty Thousand Six Hundred Seventy-Eight 4,050,678 Four Millions Fifty Thousand Six Hundred Seventy-Eight (b) 48121620 4,81,21,620 Four Crore Eighty-One Lakh

Twenty-One Thousand Six Hundred Twenty 48,121,620 Forty Eight Millions One Hundred Twenty One Thousand Six Hundred Twenty (c) 20022002 2,00,22,002 Two Crore Twenty Two Thousand Two 20,022,002 Twenty millions twenty two thousand and two (d) 246813579 24,68,13,579 Twenty Four Crore Sixty Eight Lakh Thirteen Thousand Five Hundred Seventy Nine 246,813,579 Two Hundred Forty Six Millions Eight Hundred Thirteen Thousand Five Hundred Seventy Nine (e) 345000543 34,50,00,543 345,000,543

More textbook solutions — companion p. 7

2. Write the following numbers in Indian place value notation: (a) One crore one lakh one thousand Ten
Answer: 1,01,01,010 (b) One billion one million one thousand one
Answer: 1,00,10,01,001 (c) Ten crore twenty lakh thirty thousand forty
Answer: 10,20,30,040 (d) Nine billion eighty million seven hundred thousand six hundred
Answer: 9,08,07,00,600
3. Compare and write '<', '>' or '=': (a) 30 thousand ___<___ 3 lakhs (b) 500 lakhs _____>_____ 5 million (c) 800 thousand ___<___ 8 million (d) 640 crore _____<_____ 60 billion

Page No. 10 Think and share situations where it is appropriate to (a) round up, (b) round down, (c) either rounding up or rounding down is okay and (d) when exact numbers are needed.
Answer: (a) Buying food/fruits for a group. (b) Estimating time remaining to leave for the school (catching the school bus). (c) Distance estimation between places, especially far off places. (d) Handling money in banks. Find more such situations.

More textbook solutions — companion p. 8

Page No. 11 Similarly, write the five nearest neighbours for these numbers: (a) 3,87,69,957 Nearest thousand 3,87,70,000 Nearest ten thousand 3,87,70,000 Nearest lakh 3,88,00,000 Nearest ten lakh 3,90,00,000 Nearest crore 4,00,00,000 (b) 29,05,32,481 Nearest thousand 29,05,32,000 Nearest ten thousand 29,05,30,000 Nearest lakh 29,05,00,000 Nearest ten lakh 29,10,00,000 Nearest crore 29,00,00,000

Page No. 13 From the information given in the table answer the following questions by approximation: (Refer table on P. 12–13
1. What is your general observation about this data? Share with the class.
Answer: One of the observations is – Most cities shown in the table have seen a significant rise in population from 2001 to 2011. Think of more observations.
2. What is an appropriate title for the above table?
Answer: Population of some Indian cities in 2001 and 2011.
3. How much is the population of Pune in 2011? Approximately, by how much has it increased compared to 2001?
Answer: Population of Pune in 2011 is 31,15,431. The population of Pune has approximately increased by 6 lakh.
4. Which city's population increased the most between 2001 and -2011?
Answer: Bengaluru has shown maximum increase in population which is 41,24,644.

More textbook solutions — companion p. 9

5. Are there cities whose population has almost doubled? Which are they?
Answer: Yes, they are Bengaluru, Hyderabad, Surat and Vadodara
6. By what number should we multiply Patna's population to get a number/population close to that of Mumbai?

Answer: We need to multiply Patna's population by 7.39 to get a number close to the population of Mumbai.

Page No. 14 Using the meaning of multiplication and division, can you explain why multiplying by 5 is the same as dividing by 2 and multiplying by 10?
Answer: 10 ÷ 2 is the same as 5.

Figure it out
1. Find quick ways to calculate these products: (a) 2 × 1768 × 50 = 2 × 50 × 1768 = 100 × 1768 (b) 72 × 125 [Hint: 125 = 1000 8 = 72 × 1000 8 = 72 8 × 1000 = 9 × 1000 (c) 125 × 40 × 8 × 25 = 125 × 8 × 40 × 25 = 1000 × 1000
2. Calculate these products quickly. (a) 25 × 12 = _____ 25 × 12 = 100 4 × 12 = 100 × 12 4 = 100 × 3 (b) 25 × 240 = _____ 25 × 240 = 100 4 × 240 = 100 × 240 4 = 100 × 60

More textbook solutions — companion p. 10

Page No. 15 Observe the number of digits in the two numbers being multiplied and their product in each case. Is there any connection between the numbers being multiplied and the number of digits in their product?
Answer: The maximum digits in their product will be the sum of the digits of the two numbers. The minimum digit in their product will be one less than their sum. Roxie says that the product of two 2-digit numbers can only be a 3- or a 4-digit number. Is she correct?
Answer: Yes, Roxie is correct. Should we try all possible multiplications with 2-digit numbers to tell whether Roxie's claim is true? Or is there a better way to find out?
Answer: She can just check the minimum and maximum number of digits possible. Can multiplying a 3-digit number with another 3-digit number give a 4-digit number?
Answer: No. Can multiplying a 4-digit number with a 2-digit number give a 5-digit number?
Answer: Yes.

More textbook solutions — companion p. 11

Page No. 16 5-digit × 5-digit = 9 or 10 8-digit × 3-digit = 10 or 11 12-digit × 13-digit = 24 or 25

Page No. 19 Find out if you can reach the Sun in a lifetime, if you travel 1000 kilometers every day. (You had written down the distance between the Earth and the Sun in a previous exercise.)
Answer: We cannot reach the sun in a lifetime because it takes approx. 403 years. Make necessary reasonable assumptions and answer the questions below: (a) If a single sheet of paper weighs 5 grams, could you lift one lakh sheets of paper together at the same time?
Answer: Weight of one lakh (1,00,000) sheets = 500 kg. Since a person cannot lift 500 kg so, we cannot lift one lakh sheets of paper. (b) If 250 babies are born every minute across the world, will a million babies be born in a day?
Answer: Babies born in one day = 3,60,000 babies. 3,60,000 is less than one million (10 lakh) Hence, a million babies cannot born in a day. (c) Can you count 1 million coins in a day? Assume you can count 1 coin every second.
Answer: No. Because, Coins counted per day = 86,400

More textbook solutions — companion p. 12

Figure it out

Page No. 19
1. Using all digits from 0 – 9 exactly once (the first digit cannot be 0) to create a 10-digit number, write the — (a) Largest multiple of 5

Answer: largest multiple of 5 = 9876543210 (b) Smallest even number
Answer: The smallest such even number = 1023456798
2. The number 10,30,285 in words is ten lakhs thirty thousand two hundred eighty-five, which has 42 letters. Give a 7-digit number name which has the maximum number of letters.
Answer: One such number is 77,77,777, which has 60 letters.
3. Write a 9-digit number where exchanging any two digits results in a bigger number. How many such numbers exist?
Answer: One such number is 987654312. Exchanging last two digits, we get 987654321, which is bigger number than the initial number. Try more!
4. Strike out 10 digits from the number 12345123451234512345 so that the remaining number is as large as possible.
Answer: 5534512345
6. Suppose you write down all the numbers 1, 2, 3, 4, ..., 9, 10, 11, and so on. The tenth digit you write is '1' and the eleventh digit is '0', as part of the number 10. (This question can be solved in different ways. One of the way is —) (a) What would the 1000th digit be? At which number would it occur?
Answer: Digits from 1 to 9 = 9 digits Digits from 10 to 99 = 99-10+1 = 90 numbers × 2 = 180 digits Total digits from 1 to 99 = 9 + 180 = 189 digits Remaining digits to reach 1000th digit = 1000 – 189 = 811 Number of 3-digit numbers = 811 3 = 270 full numbers + 1 digit left over. The first 3-digit number is 100 270th 3-digit number is 100 + 270 – 1 = 369

More textbook solutions — companion p. 13

Answer: The millionth digit occurs in the number 185184+1=1,85,185. (c) When would you have written the digit '5' for the 5000th time?
Answer: 13995
7. A calculator has only '+10,000' and '+100' buttons. Write an expression describing the number of button clicks to be made for the following numbers: (a) 20,800
Answer: 20,800 = (2 × 10,000) + (8 × 100) Total clicks = 2 + 8 = 10 button clicks. (b) 92,100
Answer: 92,100 = (9 × 10,000) + (21 × 100) Total clicks = 9 + 21 = 30 button clicks. (c) 1,20,500
Answer: 1,20,500 = (12 × 10,000) + (5 × 100) Total clicks = 12 + 5 = 17 button clicks. (d) 65,30,000
Answer: 65,30,000 = (653 × 10,000) Total clicks = 653 button clicks. (e) 70,25,700
Answer: 70,25,700 = (702 × 10,000) + (57 × 100) Total clicks = 759 button clicks.
8. How many lakhs make a billion?
Answer: 10,000 lakh make a billion.
9. You are given two sets of number cards numbered from 1 – 9. Place a number card in each box below to get the (a) largest possible sum (b) smallest possible difference of two resulting numbers. (a) Ans: Sum = 1,00,54,320 9 9 8 8 7 7 6 6 5 5 4 4

More textbook solutions — companion p. 14

10. You are given some number cards; 4000, 13000, 300, 70000, 150000, 20,
5. Using the cards get as close as you can to the numbers below using any operation you want. Each card can be used only once for making a particular number. (a) 1,10,000: 4000 × (20 + 5) + 13000 = 1,13,000 (b) 2,00,000:
Answer: closest estimate = 1,50,000 + 70,000 – (4000 × 5) = 2,00,000. (c) 5,80,000:
Answer: closest estimate = (1,50,000 × 4) – (4000 × 5) = 5,80,000. (d) 12,45,000:
Answer: closest estimate = (70,000 × 20) – 1,50,000 – 4,000 – (300 × 5) = 12,44,500. (e) 20,90,800:
Answer: closest estimate = (1,50,000 × 14) + 4,000 – 13,000 = 20,91,000.
11. Find out how many coins should be stacked to match the height of the Statue of Unity. Assume each coin is 1 mm thick.
Answer: Height of each coin = 1 mm Height of Statue of Unity = 180 m = 180 × 100 × 10 mm =

1, 80,000 mm. Number of coins to be stacked = 1,80,000 mm ÷ 1 mm = 1,80,000 coins.

12. Grey-headed Albatrosses have a roughly 7-feet wide wingspan. They are known to migrate across several oceans. Albatrosses can cover about 900 – 1000 km in a day. One of the longest single trips recorded is about 12,000 km. How many days would such a trip take to cross an ocean approximately? 1 1 2 2 3 3 4 9 9 8 8 7

More textbook solutions — companion p. 15

Answer: Estimated number of days if it flies 900 km/day = 12,000 ÷ 900 = 13.3 days Estimated number of days if it flies 1000 km/day = 12,000 ÷ 1000 = 12 days Hence it would take approximately 12 to 14 days for a grey headed albatross to complete a 12,000 km trip to cross an ocean.

13. A bar-tailed godwit holds the record for the longest recorded non- stop flight. It travelled 13,560 km from Alaska to Australia without stopping. Its journey started on 13 October 2022 and continued for about 11 days. Find out the approximate distance it covered every day. Find out the approximate distance it covered every hour.

Answer: Distance covered per day = 13,560 km ÷ 11 days = 1232.73 km or 1233 km. Distance covered per hour = 1233 ÷ 24 = 51.36 km or 51 km.

14. Bald eagles are known to fly as high as 4500 – 6000 m above the ground level. Mount Everest is about 8850 m high. Aeroplanes can fly as high as 10,000 – 12,800 m. How many times bigger are these heights compared to Somu's building?

Answer: Height of Somu's building = 40 m (i) Bald eagles' flight height = 4500 – 6000 m Lower estimate = 4500 ÷ 40 = 112.5 times Upper estimate = 6000 ÷ 40 = 150 times Bald eagles fly about 112 to 150 times higher than Somu's building. (ii) Mount Everest height = 8850 m Now, 8850 ÷ 40 = 221.25 Mount Everest is approximately 221 times taller than Somu's building. (iii) Aeroplanes flight height = 10,000 – 12,000 m Lower estimate = 10,000 ÷ 40 = 250 Upper estimate = 12800 ÷ 40 = 320. Airplanes fly about 250 to 320 times as high as Somu's building.

Illustrated notes: how to understand large numbers

A lakh and beyond

Key Idea

Indian grouping: the rightmost group has **three digits**, then groups have **two digits**. 1,00,000 = 1 lakh, 1,00,00,000 = 1 crore and 1,00,00,00,000 = 1 arab (one billion).

Place	Value	Example
Thousand	1,000	7,000
Ten thousand	10,000	70,000
Lakh	1,00,000	2,00,000
Ten lakh	10,00,000	30,00,000
Crore	1,00,00,000	4,00,00,000

The calculator button method

+10000	+1000	+100	+1
4 presses	6 presses	2 presses	9 presses

$$40{,}000 + 6000 + 200 + 9 = 46{,}209$$

Remember: For minimum clicks, use the largest powers-of-ten button available. Each digit tells the number of presses at that place.

Estimating instead of calculating exactly

Compare to nearby convenient values: $824 \times 25 = 824 \times \frac{100}{4} = 20{,}600$. Rounding 824 to 800 gives a quick estimate $800 \times 25 = 20{,}000$. The second is a check, not the exact product.

Worked Example

An average hall has 500 seats. About how many halls for one lakh people? $100{,}000 \div 500 = 200$ halls. If the seating capacity is an estimate, the result is approximate.

Nearest neighbours and patterns

The five consecutive neighbours around 63,215 can be $63{,}213, 63{,}214, 63{,}215, 63{,}216, 63{,}217$. Multiplying by 25 is the same as multiplying by 100 and dividing by 4. Multiplying by 125 is multiplying by 1000 and dividing by 8.

Memory Trick

10 hundreds = 1 thousand; 100 thousands = 1 lakh; 100 lakhs = 1 crore; 100 crores = 1 arab/billion.

Last-minute revision

- Put commas as XX, XX, XXX after the first three digits from the right.
- Use place-value expansion to check a large number.
- Estimate first, calculate second, and include units.
- For a counting question, write the number of events *per day* or *per second* before multiplying.

Mini-lab: place value as coloured blocks

Crore	Lakh	Thousand	Units
3	4	5	67

3,04,05,067

Read groups, not digits

The number 3,04,05,067 is **three crore four lakh five thousand sixty-seven**. Zeroes in a group still matter because they reserve a place for all the other digits.

Real-world reasoning: getting a feel for numbers

Worked Example

A bus has 50 seats. How many buses are needed for one lakh students? $100{,}000 \div 50 = 2000$ buses. This comparison helps us *visualise* a lakh.

Worked Example

At 100 km/day, the distance travelled in 10 years is $100 \times 365 \times 10 = 365{,}000$ km. Compared with 384,400 km to the Moon (the chapter's approximate figure), that is still short by 19,400 km.

Quick self-check (with answers)

1. How many tens make 75,000? **Answer:** 7500.
2. 6 lakh + 45 thousand + 72 = ? **Answer:** 6,45,072.
3. Find 208×25 mentally. **Answer:** $(208 \div 4) \times 100 = 5200$.
4. Estimate 998×51. **A good estimate:** $1000 \times 50 \approx 50{,}000$.
5. Seventeen persons enter a hall every minute. How many in an hour? **Answer:** $17 \times 60 = 1020$.
6. True or false: 1 billion is 1000 crore. **False:** it is 100 crore.

Memory Trick

When answering a large-number problem, remember **place value, operation, unit, check**. A simple estimate catches many mistakes.